

# Matius Robert

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Singapore

## EXPERIENCE

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- **Doctoral Research (PhD)** Aug 2025 - Current  
*Duke-NUS Medical School* Singapore
  - Integrated Biology and Medicine (IBM) Programme
- **Research Assistant** Sept 2021 - May 2025  
*NUHS Centre for Healthy Longevity* Singapore
  - Led and drive research and collaborations on projects investigating the senescence secretome and extrachromosomal DNA (ecDNA) in cancer progression.
  - Awarded research funding via Young Investigator Grant from the Healthy Longevity Research Program (HLTRP) for new pilot study on extrachromosomal DNA.
  - Designed and executed analyses of clinical information and differential gene expression analysis on publicly available databases such as TCGA, TCIA, and single-cell atlases.
  - Techniques implemented include time-lapse confocal microscopy, karyotyping, fluorescence in-situ hybridization (FISH), and DNA content profiling via flow cytometry.
- **Undergraduate Researcher** Aug 2020 - Jun 2021  
*Department of Physiology, National University of Singapore* Singapore
  - Designed and executed a loss-of-function study to investigate the role of a histone deacetylase for specific targeting of glioblastoma cancer stem cells.
  - Techniques implemented include site-directed mutagenesis, limiting dilution assay, and in-vitro cancer phenotypic assays.
- **Undergraduate Researcher** June 2019 - April 2020  
*Department of Biological Sciences, National University of Singapore* Singapore
  - Carried out phenotypic & molecular characterization of a plant disease resistance protein in the context of stomatal development in Arabidopsis.
  - Implemented protein-protein interaction assays such as yeast two hybrid and bimolecular fluorescence complementation for validation of high-throughput screening hits.

## EDUCATION

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- **Doctor of Philosophy** 2025 - 2029  
*Duke-NUS Medical School* Singapore
  - Recipient of Duke-NUS PhD Fellowship
- **Bachelor of Science (Hons.)** 2017 - 2021  
*National University of Singapore* Singapore
  - Recipient of the Student Life Award (Community Engagement - Merit)
  - Societies: NUSSU Committee for Information Technology; NUS Angklung Ensemble; NUS Cultural Activities Club

## PUBLICATIONS

J=Journal, P=Pre-print

- [J.1] **M. Robert**, B.K. Kennedy, K.C. Crasta. (2024). **Therapy-induced senescence through the redox lens**. *Redox Biology*, Vol. 74, Article 103228. DOI: 10.1016/j.redox.2024.103228
- [J.2] Y.H Chuah, E.X.Y Tay, O.V. Grinchuk, J. Yoon, J. Feng, S. Kannan, **M. Robert**, R. Jakhar, Y. Liang, B.W.L Lee, L.C. Wang, Y.T. Lim, T. Zhao, R.M. Sobota, G. Lu, B.C. Low, K.C. Crasta, C.S. Verma, Z. Lin, D.S.T Ong. (2023). **CAMK2D serves as a molecular scaffold for RNF8-MAD2 complex to induce mitotic checkpoint in glioma**. *Cell Death & Differentiation*, Vol. 30, pp. 1973-1987. DOI: 10.1038/s41418-023-01192-3
- [J.3] **M. Robert**, K.C. Crasta. (2022). **Breaking the vicious circle: extrachromosomal circular DNA as an emerging player in tumour evolution**. *Seminars in Cell & Developmental Biology*, Vol. 123, pp. 140-150. DOI: 10.1016/j.semcdb.2021.11.015
- [P.1] **M. Robert\***, R. Jakhar\*, B.V. Au, G.W.L. Eng, M. Wei, A.K. Gao, Y.H. Chuah, K. Sachaphibulkij, I. Bonne, K.J. Lim, I. Sinha, D.B.L Teh, L.H.K. Lim, P. Kumar, N.K. Verma, N.S.K. Sze, E.H. Lim, D.S.T. Ong, J.K. Cheong, K. Itahana, J.E. Connolly, K.C. Crasta. (2024). **Tumour-derived extracellular vesicles within the therapy-induced senescent secretome distinctly suppress breast cancer via DKK1-mediated inflammatory response**. *BioRxiv*. DOI: 10.1101/2024.03.29.586905

## CONFERENCES & TALKS

O=Oral, PO=Poster

- [O.1] **Matius Robert\***, Rekha Jakhar\*, Bijin Veonice Au, Gracie Wee Ling Eng, Alvin Kunyao Gao, You Heng Chuah, Karishma Sachaphibulkij, Lina Hsiu Kim Lim, Derrick Sek Tong Ong, Elaine Hsuen Lim, Yoko Itahana, Koji Itahana, Jit Kong Cheong, John Edward Connolly, Karen Carmelina Crasta. (2024). **Extracellular vesicles within a tumourigenic senescent tumour secretome are able to confer antitumour immunity**. In *N2CR Early Cancer Intervention Symposium*, 13-14 May 2024, MD6 Building, National University of Singapore, Singapore.
- [PO.1] **Matius Robert\***, Rekha Jakhar\*, Bijin Veonice Au, Gracie Wee Ling Eng, Alvin Kunyao Gao, You Heng Chuah, Karishma Sachaphibulkij, Lina Hsiu Kim Lim, Derrick Sek Tong Ong, Elaine Hsuen Lim, Yoko Itahana, Koji Itahana, Jit Kong Cheong, John Edward Connolly, Karen Carmelina Crasta. (2023). **Small Extracellular Vesicles from Therapy-induced Senescent TNBC Elicit Paracrine Anti-tumour Effects**. In *Frontiers in Cancer Science*, 6-8 November 2023, University Cultural Centre, National University of Singapore, Singapore.
- [PO.2] **Matius Robert\***, Rekha Jakhar\*, Karen Carmelina Crasta. (2022). **Role of small extracellular vesicles in mediating paracrine activities of therapy-induced senescent breast cancer**. In *Longevity Science Singapore Conference*, 7-8 September 2022, NUHS Tower Block, National University of Singapore, Singapore.

## SKILLS

- **Programming Languages:** R, Python, Bash
- **Version Control & Deployment:** Git, Docker
- **Specialized Area & Interest:** Senescence, Biogerontology, Cancer biology, Genomic instability, Multiomics
- **Languages:** English (Bilingual), Bahasa Indonesia (Native)

## HONORS AND AWARDS

- **Duke-NUS PhD Fellowship** Aug 2025  
*Duke-NUS Medical School* 
  - Duke-NUS PhD Fellowship is a merit scholarship awarded to graduate students under the Duke-NUS PhD programme. The scholarship provides monthly stipend, travel grant, tuition fee waiver, and other miscellaneous student fee subsidy.
- **HLTRP Young Investigator Grant** Nov 2024  
*Healthy Longevity Translational Research Programme* 
  - HLTRP Young Investigator Grant aims to support innovation amongst junior scientists. The grant awards selected research proposals with emphasis on blue-sky research and novelty in the field of healthy longevity.
- **The Student Life Awards (Community Engagement - Merit)** Oct 2019  
*NUS Office of Student Affairs* 
  - The Student Life Awards is an annual event organized by the Office of Student Affairs (OSA) that aims to recognize exceptional students for their noteworthy contributions to the University outside of academia.

## VOLUNTEER EXPERIENCE

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### • Coordinator & Volunteer

May 2022 - Jan 2025

*Healthy Longevity Webinar Series*



- The webinar series aims to raise awareness of healthy ageing and invite leading scientists, businessmen, and policymakers to share perspectives on ageing research. It reaches an international audience of scientists and laymen with a participation of 350 audiences on average for each episode.
- As volunteer, I provide concise 5-minute presentations catered to the public on recent findings in the field of geroscience as well as on ongoing research projects in CHL.
- As coordinator, I am responsible for the recruitment, preparation, scheduling, and general coordination of the internal presenters and moderators to ensure smooth production workflow and high-quality weekly episodes.

### • Volunteer

Sept 2022

*Longevity Science Singapore Conference*



- The Longevity Science Singapore Conference 2022 is a full-day medical scientific conference catered for healthcare and academic professionals. It invites leading international and local experts on healthy longevity to discuss on the latest strategies and discoveries in longevity interventions.
- As volunteer, I provided hospitality for guest speakers and VIPs, as well as performed logistical *ad-hoc* duties for the conference preparation.

### • Logistics Director

Feb 2018 - Oct 2018

*NUANSA Cultural Production*



- NUANSA 2018 is a cultural production about self-belief, put together by 105 students from different nationalities. The production was held at Resorts World Sentosa Theatre and garnered about 700 audiences, with part of its proceeds donated to earthquake victims in Palu, Indonesia.
- As logistics director, I lead and managed the logistics team to provide timely logistical needs and management of inventories for the whole committee. These includes venue arrangements, transportation, procurements, as well as manpower needs.

## REFERENCES

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### 1. Karen Carmelina Crasta

Research Assistant Professor, Department of Physiology  
National University of Singapore  
Email: karenocrasta@nus.edu.sg  
*Relationship: Project Supervisor, Reporting Manager*

### 2. Derrick Ong Sek Tong

Assistant Professor, Department of Physiology  
National University of Singapore  
Email: phsostd@nus.edu.sg  
*Relationship: Bachelor's Thesis Advisor*